

Vendor Performance Data Analysis

End-to-End Data Engineering, Analytics & Business Intelligence Project

Project Objective

- Develop an automated vendor performance monitoring system using Python, SQL, and Power BI.
- Build a scalable ETL pipeline to process large volumes of vendor sales data.
- Analyze vendor profitability, purchasing efficiency, and inventory performance.
- Provide business stakeholders with an interactive dashboard for data-driven decision making.

Tech Stack

- **Database:** SQL
- **Programming:** Python
- **Libraries:** Pandas, NumPy, SQLAlchemy, Matplotlib, Seaborn
- **Visualization:** Power BI
- **Architecture:** ETL Pipeline + Business Intelligence Dashboard

Business Problem Statement

Problem Statement

Organizations purchasing products from multiple vendors face several challenges:

Challenges

- Large amount of raw sales and inventory data stored in databases.
- Difficult to identify high-performing and low-performing vendors.
- Lack of visibility into unsold inventory and blocked capital.
- Manual reporting consumes significant business time.
- Decision-makers cannot quickly identify purchasing inefficiencies.

Business Goal

Design a complete analytics pipeline that:

- Automatically extracts data from database.
- Cleans and transforms raw datasets.
- Generates vendor-level performance metrics.
- Identifies profit opportunities and inventory risks.
- Creates an interactive dashboard for strategic decision making.

Project Workflow (ETL Pipeline)

End-to-End Data Pipeline

01

Step 1 – Data Extraction

- Connect SQL Database using SQLAlchemy Engine.
- Load large-volume transactional data.

02

Step 2 – Ingestion Pipeline (*ingest.py*)

- Database Connection
- Read Raw Tables
- Filter Required Records
- Upload Processed Data Back to SQL Database

03

Step 3 – Transformation Pipeline (*get_vendor_summary.py*)

- Handle Missing Values
- Remove Duplicate Records
- Feature Engineering
- Vendor-wise Aggregation
- KPI Calculation
- Profit Margin Calculation

04

Step 4 – Data Analysis

- Exploratory Data Analysis using Python
- Vendor Comparison
- Brand Analysis
- Profitability Analysis

05

Step 5 – Dashboard Development

- Interactive Power BI Dashboard
- Business KPIs
- Vendor Performance Monitoring

Business Questions Solved

Key Business Questions

The project answers important business questions such as:

1. Which vendors generate the highest sales?
2. Which brands contribute maximum revenue?
3. Which vendors are underperforming?
4. Which brands have poor profitability?
5. What is the company's total profit?
6. What is the overall profit margin?
7. How much capital is locked in unsold inventory?
8. Which vendors contribute most to total sales?
9. Are high sales always associated with high profit?
10. Which vendors should receive higher purchase priority?



Dashboard KPIs

Business Performance Summary

Overall Business Metrics

\$441.41M

Total Sales

\$307.34M

Total Purchase

\$134.07M

Gross Profit

38.7%

Profit Margin

\$2.71M

Unsold Capital

Dashboard Features

- Purchase Contribution Analysis
- Top Vendors by Sales
- Top Brands by Sales
- Low Performing Vendors
- Low Performing Brands
- Profit Margin Distribution

These KPIs provide management with an instant overview of overall business health.

Vendor Performance Dashboard

441.41M

Total Sales (\$)

307.34M

Total Purchase (\$)

134.07M

Gross Profit (\$)

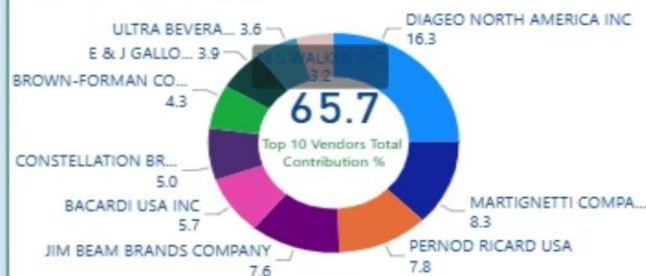
38.7

Profit Margin (%)

2.71M

Unsold Capital (\$)

Purchase Contribution %



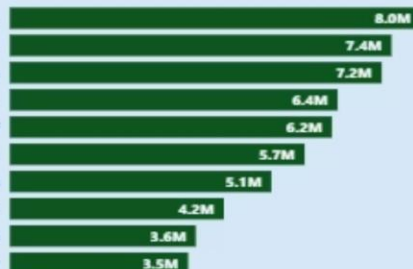
Top Vendors By Sales

DIAGEO NORT...
MARTIGNETTI ...
PERNOD RICA...
JIM BEAM BRA...
BACARDI USA I...
CONSTELLATI...
E & J GALLO W...
BROWN-FORM...
ULTRA BEVERA...
M S WALKER I...

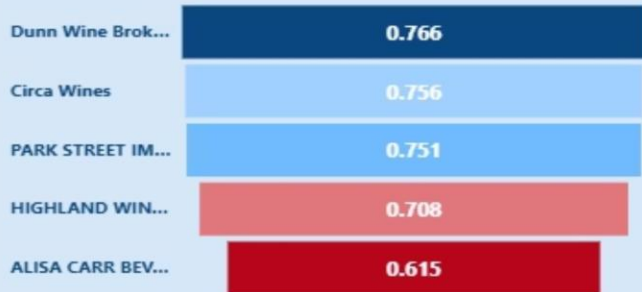


Top Brands By Sales

Grey Goose Vodka
Capt Morgan Spi...
Absolut 80 Proof
Jameson Irish Wh...
Ketel One Vodka
Baileys Irish Cream
Kahlua
Tanqueray



Low Performing Vendors



Low Performing Brands

◆ No ◆ Yes



Key Insights from Dashboard

Major Business Insights

Revenue Insights

- Business generated **\$441M+** in total sales.
- Gross profit exceeded **\$134M**, indicating strong profitability.
- Overall profit margin of **38.7%** reflects efficient pricing and purchasing strategies.

Vendor Insights

- A small number of vendors contribute the majority of total sales.
- Top vendors significantly outperform the remaining supplier base.
- Some vendors consistently deliver poor performance and require evaluation.

Brand Insights

- Premium brands dominate revenue generation.
- Several brands contribute minimal sales despite inventory investment.

Inventory Insight

- **\$2.71M** remains blocked as unsold inventory.
- Indicates opportunities for inventory optimization and better purchasing decisions.

Business Recommendations

Recommendations

Vendor Optimization

- Strengthen partnerships with top-performing vendors.
- Renegotiate contracts with low-performing vendors.
- Consider replacing consistently underperforming suppliers.

Inventory Management

- Reduce procurement of slow-moving products.
- Improve demand forecasting.
- Introduce inventory monitoring alerts.

Profitability Improvement

- Focus marketing on high-margin brands.
- Analyze low-margin products for pricing optimization.
- Monitor vendor performance monthly.

Decision Support

- Use the Power BI dashboard for executive reporting.
- Automate ETL pipeline for real-time business monitoring.

Project Outcome & Business Impact

Project Outcome

Technical Achievements

- Built a scalable ETL pipeline using Python.
- Automated database extraction, transformation, and loading.
- Developed reusable data processing modules.
- Created an interactive Power BI dashboard for business users.

Business Impact

- Reduced manual reporting effort.
- Faster decision-making through centralized dashboards.
- Improved visibility into vendor and brand performance.
- Enabled proactive inventory management.
- Supported data-driven procurement and profitability strategies.

Conclusion

This project demonstrates how **Data Engineering + Data Analysis + Business Intelligence** can transform raw transactional data into actionable business insights, helping organizations optimize vendor performance, improve profitability, and make strategic purchasing decisions.